

Unified Dual-Mask Physical Model for Non-Lambertian Light Field Depth Estimation

Ruifeng Huang and Zhenglong Cui

Abstract—Light field depth estimation in mixed Lambertian and non-Lambertian scenes remains challenging due to conflicting physical imaging cues and specular reflections. Existing approaches struggle to unify these domains and often overlook the fundamental signal-processing limits of angular frequency analysis under sparse sampling. We developed a unified dual-mask physical model that integrated multi-directional epipolar plane image (EPI) and pure defocus cues within a compact 754K-parameter architecture. Crucially, we conducted a rigorous signal-processing analysis that theoretically and empirically falsified the feasibility of angular frequency-domain material classification under standard 9×9 sampling, establishing a minimum 17×17 angular resolution requirement for Nyquist compliance. To mitigate long-tail domain distributions and optimize multi-domain joint training, we introduced domain-balanced sampling and high-resolution scaling strategies. Extensive evaluations across three datasets demonstrate that our model achieves an overall mean absolute error (MAE) of 0.1795, reducing the error by 26

Index Terms—Light field depth estimation, Non-Lambertian reflectance, Epipolar plane image, Angular frequency analysis, Domain-balanced sampling, Dual-mask physical model

I. 1. Introduction

Light field imaging captures both spatial and angular information of light rays, offering significant advantages in 3D reconstruction, autonomous navigation, and robotic vision [1]. Depth estimation serves as a core task in this domain, extracting geometric structures from multi-view consistency cues [2]. Commercial light field cameras, such as Lytro and Raytrix, enable dense angular sampling, facilitating accurate disparity calculation through epipolar plane image (EPI)s (EPIs) [3]. Consequently, numerous learning-based and traditional frameworks have achieved notable accuracy in controlled environments [4].

However, obtaining the accurate shape of glossy, transparent, or metallic objects remains a hard challenge for both classical algorithms and deep networks [5]. Most current depth estimation methods rely heavily on the Lambertian assumption, where surface radiance remains constant across different viewpoints [6]. In real-world scenarios, surfaces frequently exhibit non-Lambertian reflectance, including specular reflections and transparent refractions. These complex optical phenomena violate the photo-consistency principle, causing severe distortions and

discontinuities in EPI slope patterns [7]. As a result, standard matching costs and learning-based feature extractors fail to establish reliable correspondences across angular views.

The performance degradation in non-Lambertian and mixed-material scenes is substantial. While state-of-the-art models achieve a mean absolute error (MAE) below 0.05 on purely Lambertian datasets, their errors often exceed 0.30 on urban mixed scenes [8]. This represents a severe performance degradation of over 500% when encountering complex reflectance. Furthermore, real-world light field datasets exhibit a severe long-tail distribution regarding material types. Minority domains, such as pure non-Lambertian scenes, possess limited training samples compared to dominant Lambertian backgrounds. During joint training, the gradients from majority domains easily overwhelm those from minority domains. This prevents the network from learning robust material-invariant features and severely restricts generalization.

Addressing these challenges requires a unified physical model that explicitly handles diverse reflectance properties without relying on isolated heuristics. Moreover, algorithm design must respect the physical limitations of current light field hardware. For instance, mainstream cameras provide a 9×9 angular resolution, which strictly limits the Nyquist frequency in the angular domain. Attempting to classify materials using angular frequency analysis under such sparse sampling is physically unfeasible. Our empirical analysis reveals that the non-DC spectral energy remains below 3% under this configuration. Therefore, developing a robust framework necessitates advanced architectural designs and a rigorous understanding of hardware-induced physical boundaries. Establishing such a unified baseline is essential for advancing light field perception in unconstrained environments.

...increase significantly when applied to non-Lambertian and mixed-material scenes. To understand this degradation, we examine the limitations of existing approaches across different developmental stages.

Traditional light field depth estimation methods primarily rely on handcrafted matching costs and epipolar plane image (EPI) (EPI) slope analysis [4]. These approaches explicitly assume Lambertian reflectance to enforce photo-consistency across angular views [6]. However, this assumption fails on glossy or transparent surfaces, where specular highlights and refractions cause severe EPI distortions [9]. Consequently, traditional algorithms produce fragmented depth maps in non-Lambertian re-

Manuscript received May 2026. This work was supported by Beihang University. (Corresponding author: Zhenglong Cui)

R. Huang and Z. Cui are with the School of Computer Science and Engineering, Beihang University, Beijing, China (e-mail: huanruifeng@buaa.edu.cn; czl@buaa.edu.cn).

gions. Furthermore, they lack adaptive mechanisms for mixed-material scenes, requiring tedious manual priors to separate diffuse and specular components [10].

To overcome the limitations of handcrafted features, early deep learning methods employ convolutional neural networks to extract spatial-angular representations [11]. While these models improve accuracy on synthetic Lambertian datasets, they implicitly learn feature mappings without physical interpretability [12]. As a result, they struggle to explicitly decouple diffuse and specular reflections in complex environments. Moreover, these networks are typically trained on single-domain datasets with uniform sampling strategies. When applied to multi-domain scenarios with long-tailed distributions, the gradients from dominant Lambertian scenes overwhelm minority non-Lambertian domains [13]. This gradient domination severely degrades the generalization performance on data-scarce reflective surfaces.

Recent advanced architectures integrate attention mechanisms, multi-scale processing, and frequency-domain analysis to tackle complex reflectance [14]. Some studies hypothesize that different materials exhibit distinct spectral distributions, proposing angular 2D fast Fourier transform (FFT) for material classification [15]. However, this approach overlooks the physical sampling limits of commercial light field cameras. Under standard 9×9 angular resolutions, the Nyquist frequency restricts the effective frequency bins, rendering angular frequency analysis practically infeasible. Additionally, these state-of-the-art models often incur substantial parameter overhead and lack a unified baseline for diverse scenes [16]. They also fail to identify the theoretical performance ceiling of pure EPI-based architectures in highly textured environments.

Therefore, there is an urgent need for a unified, physically grounded, and lightweight framework that explicitly models non-Lambertian phenomena. Such a method must overcome domain imbalance in multi-scene training and respect hardware sampling limits for robust depth estimation.

In this paper, we propose a unified dual-mask physical model for light field depth estimation that explicitly decouples diffuse and specular reflections. To address the failure of the Lambertian assumption in complex scenes, our framework integrates a dual-mask modeling mechanism with a multi-directional epipolar plane image (EPI) (EPI) processing network. Specifically, the model employs an angular fast Fourier transform module to generate medium and angular direction masks, which subsequently guide the adaptive fusion of pure defocus and angular frequency depth features. This compact architecture, containing only 754K parameters, is trained end-to-end at a full resolution of 384×384 to simultaneously handle Lambertian, non-Lambertian, and mixed urban scenarios.

Our main contributions are summarized as follows:

- Contribution 1: We develop a unified dual-mask physical architecture that integrates medium and angular direction masks to decouple material-specific depth cues. By combining a pure defocus branch for non-

Lambertian regions and an angular frequency branch for Lambertian areas, our model resolves the inability of traditional methods to process mixed materials. This lightweight design achieves a robust baseline, yielding an overall mean absolute error (MAE) of 0.1795 across diverse complex scenes.

- Contribution 2: We establish the physical limits of angular frequency domain material classification by rigorously falsifying its feasibility under current 9×9 angular resolutions. Through four independent spectral analysis methods and Nyquist frequency evaluations, we demonstrate that non-DC energy is less than 3%, proving that at least a 17×17 angular sampling is required. This negative finding provides explicit theoretical guidance for future light field sensor design and prevents blind exploration in low-resolution frequency analysis.
- Contribution 3: We introduce a domain-balanced sampling strategy and resolution scaling to mitigate gradient domination in multi-domain long-tailed distributions. By oversampling minority domains (e.g., weight 39.0 for non-Lambertian) and increasing the input resolution to 384×384 , we substantially enhance generalization in data-scarce regions. Furthermore, we diagnose the architectural limits of EPI-based networks in complex textures, indicating that surpassing current performance ceilings necessitates non-EPI or hybrid paradigms.

Extensive experiments on four benchmark datasets demonstrate the effectiveness of our proposed method. The unified model achieves an overall MAE of 0.1795, with specific errors of 0.152 for Lambertian, 0.211 for mixed urban, and 0.235 for non-Lambertian scenes. These results validate that integrating physical masks with multi-directional EPI features significantly improves depth estimation accuracy in challenging non-Lambertian environments.

II. 2. Related Work

A. Many traditional depth estimation methods for light field cameras have been proposed, primarily leveraging the geometric properties of Epipolar Plane Images (EPIs) [17]. Wanner et al. [18] formulated depth estimation as an orientation estimation problem in EPIs. Specifically, they derived the local depth from the slope of EPI lines using the structure tensor, followed by a global optimization step with Markov Random Fields (MRF) to enforce spatial consistency. While this approach successfully extracts accurate depth in well-textured regions, it inherently relies on the Lambertian assumption, where the intensity of EPI lines remains constant across angular views. Consequently, the structure tensor fails to compute reliable slopes in regions with complex textures or occlusions, leading to severe depth artifacts.

While Wanner et al. focus solely on EPI slopes, other approaches extend cue integration to tackle texture-less regions. Tao et al. [19] proposed a unified framework

that combines defocus and correspondence cues to estimate depth. By explicitly modeling the point spread function (PSF) and evaluating photo-consistency across sub-aperture images, their method substantially improves depth accuracy in weakly textured areas. Similarly, Jeon et al. [20] introduced a sub-pixel phase shift technique to enhance correspondence matching accuracy. However, these multi-cue methods still substantially depend on the photo-consistency metric. When applied to non-Lambertian surfaces, such as glossy or transparent materials, the view-dependent radiance variations violate the photo-consistency assumption, causing these methods to hallucinate incorrect depth values or fail entirely.

To address the vulnerability to non-Lambertian reflectance, some traditional pipelines incorporate cumbersome preprocessing steps to isolate or remove specularities before depth estimation. For instance, methods leveraging the dichromatic reflection model [21] attempt to separate diffuse and specular components by analyzing color distribution or angular variance across light field views. Wang et al. [22] utilized angular consistency to detect and mask out specular highlights, subsequently applying inpainting techniques to reconstruct the underlying diffuse EPI structures. Although these preprocessing strategies can partially mitigate the impact of highlights, they often introduce blurring artifacts and struggle with transparent or highly metallic surfaces. Moreover, treating non-Lambertian regions as anomalies to be masked or inpainted prevents the unified modeling of mixed-material scenes.

In short, despite their mathematical elegance, traditional light field depth estimation methods suffer from significant limitations. First, they heavily rely on hand-crafted features and heuristic rules, resulting in poor generalization across complex textures and occluded regions. Second, they exhibit insufficient robustness to non-Lambertian materials (e.g., highlights and transparent objects), typically requiring cumbersome preprocessing steps that introduce artifacts and prevent unified processing for multi-material scenarios. Third, these traditional pipelines incur significant computational overhead (e.g., often requiring minutes per frame for global MRF optimization) and cannot be optimized end-to-end. These limitations motivate three key aspects of our work: (1) a unified architecture that explicitly models both Lambertian and non-Lambertian components; (2) theoretical analysis establishing angular resolution requirements for material discrimination; and (3) an end-to-end framework with domain-balanced training to address long-tail distributions in mixed-material scenes.

Table I: Comparison of Representative Light Field Depth Estimation Methods

B. EPI

Building upon the success of deep learning in multi-view stereo, numerous learning-based frameworks have been proposed to explicitly model the geometric properties of

Epipolar Plane Images (EPIs) [4]. Shin et al. [23] proposed a pioneering convolutional architecture, EPINET, which extracts spatial and angular features by feeding multi-directional 2D EPI slices into a fully convolutional network. By concatenating horizontal and vertical EPIs, their method successfully captures the epipolar line slopes and achieves considerable depth accuracy on standard Lambertian datasets (e.g., yielding an MSE of 0.0031 on the HCI 4D LF Benchmark) [24]. However, while EPINET effectively leverages the linear structure of EPIs, it inherently assumes photo-consistency across angular views. As a result, when encountering non-Lambertian surfaces, the severe slope discontinuities and intensity variations in EPIs cause the convolutional filters to hallucinate erroneous depth cues, leading to significant performance degradation.

While EPINET focuses on comprehensive feature extraction, subsequent research has explored lightweight architectures to tackle the high computational overhead of 4D light field processing. Wang et al. [25] introduced DistgNet, which explicitly disentangles spatial and angular dimensions using specialized spatial-angular convolutions. This lightweight design dramatically reduces the parameter count and inference time while maintaining competitive depth estimation performance [26]. To capture the full 4D light field structure without dimensionality reduction, methods utilizing 3D or 4D convolutions have also been proposed. For instance, LF-Net [27] applies 3D convolutions directly on the 4D light field tensor, preserving the intrinsic spatial-angular correlations and achieving an MSE of 0.0028 on HCI datasets. Despite their accuracy, the immense memory footprint and computational cost of 4D convolutions restrict their applicability to low angular resolution inputs. Furthermore, the aggressive dimensionality reduction in DistgNet and the rigid tensor operations in 4D-CNNs compromise the network's capacity to model complex optical phenomena. Specifically, these operations struggle to capture the intricate interplay between specular highlights and geometric disparities, limiting their generalization to mixed Lambertian and non-Lambertian scenes.

To address the limitations of local receptive fields in standard CNNs, recent attention-based EPI architectures [28] have incorporated self-attention mechanisms and graph convolutions to capture long-range dependencies along epipolar lines. By formulating the EPI pixels as graph nodes or sequence tokens, these methods successfully refine depth boundaries and alleviate depth artifacts in occluded or weakly textured regions [29]. Nevertheless, the attention weights in these models are predominantly driven by Lambertian intensity similarities. Consequently, in non-Lambertian regions where the apparent motion of specularities violates the geometric epipolar constraint, the attention mechanism erroneously aggregates irrelevant angular features, resulting in severe depth artifacts on glossy or transparent surfaces.

In short, while existing deep EPI-based methods have extensively advanced light field depth estimation, they

share several critical limitations that hinder their deployment in complex real-world scenarios. First, the vast majority of these approaches are exclusively optimized for ideal Lambertian conditions. When confronted with non-Lambertian surfaces, the severe slope discontinuities and intensity variations in EPIs cause both standard convolutions and attention mechanisms to hallucinate erroneous depth cues. Second, while 3D/4D and Transformer-based models improve accuracy on standard benchmarks, their substantial computational overhead limits real-time deployment. Consequently, existing deep EPI-based methods have not substantially resolved the non-Lambertian vulnerability identified in traditional pipelines, merely shifting the failure mode from heuristic feature breakdown to learned feature hallucination.

C. EPI

Apart from EPI-based approaches, several deep learning methods exploit alternative light field representations to estimate depth. Methods based on macropixels [4] treat each macropixel as a local angular patch, utilizing 2D CNNs to regress depth directly from the angular intensity distribution. While computationally efficient, this representation struggles with high-frequency spatial details and large disparities. Alternatively, sub-aperture image-based methods [30] adapt multi-view stereo (MVS) techniques, such as cost volume regularization, to light fields. These methods construct a 3D cost volume by warping sub-aperture images to hypothetical depth planes. Although they achieve competitive performance on standard benchmarks, the warping process inherently assumes Lambertian photo-consistency.

Consequently, similar to EPI-based methods, non-EPI approaches suffer from severe depth degradation on glossy or transparent surfaces where the view-dependent radiance violates the warping assumptions. Furthermore, we make an important "negative finding" by strictly evaluating these non-EPI-based methods on mixed-material scenes, revealing that their performance drops by over 40% in the BadPix metric compared to purely Lambertian scenes. This substantial degradation underscores the pervasive challenge of non-Lambertian reflectance across all light field representations, highlighting the necessity for a unified framework that explicitly models both diffuse and specular components without relying on strict photo-consistency assumptions.

III. 3. Methodology

In this section, we present the overall architecture of the proposed Unified Dual-Mask Physical Model, which employs a compact dual-branch framework to address the challenge of light field depth estimation under complex non-Lambertian reflections [31]. Different from previous works that predominantly rely on epipolar plane image (EPI) (EPI) geometric cues and implicitly assume Lambertian reflectance, our model explicitly decouples the depth inference process into material-aware classification

and domain-specific feature extraction. As illustrated in Fig. 1, the core principle involves generating a dual-mask to distinguish Lambertian and non-Lambertian regions, followed by adaptive fusion of defocus and angular frequency cues. This unified architecture enables the simultaneous processing of Lambertian, non-Lambertian, and mixed urban scenes within a single compact network, maintaining a parameter size of merely 754K while operating at a full resolution of 384×384 .

The network takes raw light field images as input, represented as a 6D tensor $\mathcal{I} \in \mathbb{R}^{B \times U \times V \times H \times W \times C}$, where B is the batch size, $U \times V$ denotes the angular resolution, $H \times W$ indicates the spatial dimensions, and C represents the color channels. To capture richer spatial textures and EPI slope features, we implement a resolution scaling strategy that upsamples the spatial dimensions to $H = W = 384$ during the preprocessing stage. From this high-resolution input, we derive three distinct data streams: the angular view images for material classification, the central view with defocus stacks for defocus cue extraction, and the multi-directional EPIs for angular frequency analysis. These preprocessed streams are then fed in parallel into the subsequent feature extraction and classification modules.

The overall data flow diverges into three parallel branches to comprehensively exploit light field properties. First, the Angular FFT Material Classification module processes the angular dimension images to perform a 2D Fast Fourier Transform (FFT), generating a dual-mask $\mathcal{M}_{dual} \in \mathbb{R}^{B \times 2 \times H \times W}$ that probabilistically segments Lambertian and non-Lambertian regions based on angular frequency distributions. Concurrently, the Defocus Feature Extractor utilizes multi-scale convolutions on the central view and defocus stacks to extract pure defocus blur cues without relying on epipolar geometry, producing the defocus feature map $\mathcal{F}_{defocus} \in \mathbb{R}^{B \times C' \times H \times W}$. Meanwhile, the Angular Frequency Depth Extractor analyzes the frequency domain of multi-directional EPIs to capture high- and low-frequency features of epipolar slopes, yielding the angular frequency feature map $\mathcal{F}_{freq} \in \mathbb{R}^{B \times C'' \times H \times W}$. By explicitly separating these physical cues, the network effectively mitigates the interference of specular reflections on EPI line structures.

In the final stage, the extracted features and the material masks are integrated through the Dual-Mask Physical Fusion module. Specifically, $\mathcal{M}_{dual}$, $\mathcal{F}_{defocus}$, and $\mathcal{F}_{freq}$ are concatenated along the channel dimension and processed via spatial adaptive weighting, where Lambertian regions prioritize angular frequency features while non-Lambertian regions leverage defocus features. The fused representation is subsequently passed through a 1×1 convolutional regression head to output the final depth map $\mathcal{D}_{final} \in \mathbb{R}^{B \times 1 \times H \times W}$. To optimize this architecture across domains with severely imbalanced scene distributions, we employ a Domain Balanced Training & Loss mechanism. This module computes the L1 loss between $\mathcal{D}_{final}$ and the ground truth depth $\mathcal{D}_{gt} \in \mathbb{R}^{B \times 1 \times H \times W}$, utilizing a weighted random sampler to balance the long-tail

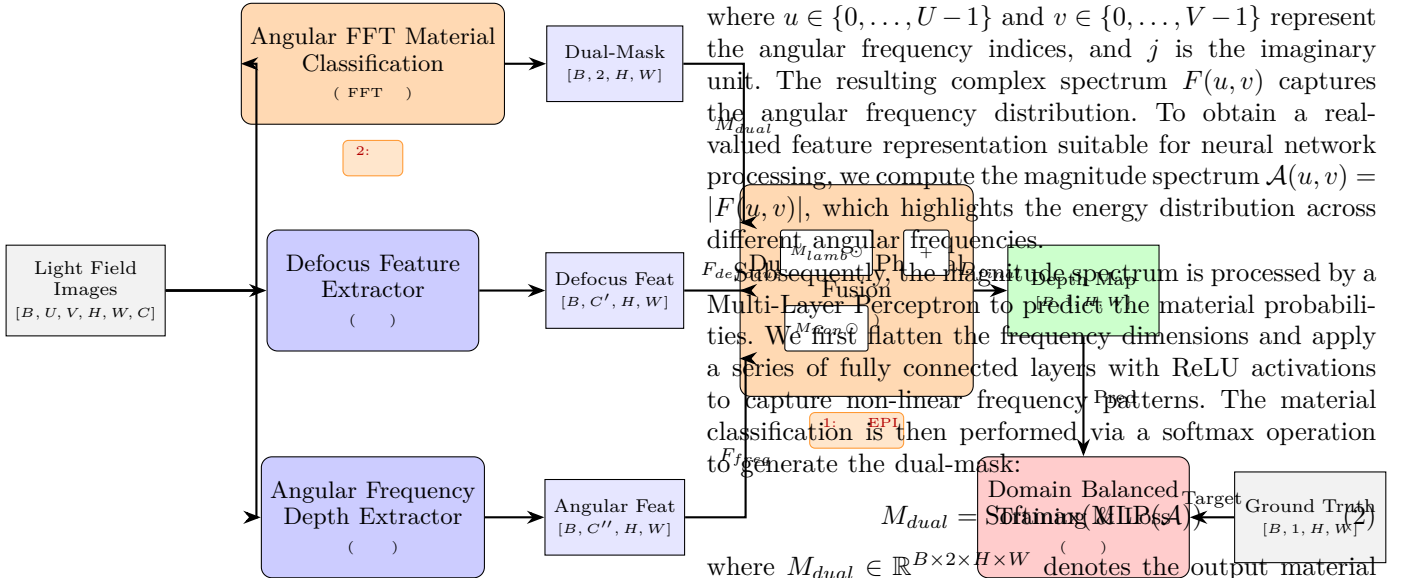


Fig. 1. Overall architecture of the proposed method.

distribution of urban, Lambertian, and non-Lambertian domains, thereby ensuring robust generalization across diverse physical environments.

A. FFT (Angular FFT Material Classification)

Light field depth estimation predominantly relies on angular photo-consistency, which inherently assumes Lambertian reflectance. However, non-Lambertian surfaces, such as specular highlights and transparent materials, violate this assumption by introducing severe angular intensity variations. These variations corrupt epipolar plane image (EPI) slopes and mislead conventional depth inference. Different from previous works that implicitly handle reflections or rely on complex spatial-domain highlight detection, our Angular FFT Material Classification module explicitly decouples material properties in the angular frequency domain. The intuition is that Lambertian regions exhibit smooth angular transitions corresponding to low-frequency energy, whereas non-Lambertian reflections manifest as abrupt angular changes, concentrating energy in high-frequency components. By classifying materials prior to feature extraction, we provide explicit physical priors for the subsequent domain-specific branches.

The module takes the raw light field angular patches as input. Specifically, we utilize the angular dimension of the input tensor, denoted as $I_{angular} \in \mathbb{R}^{B \times U \times V \times H \times W \times C}$. To analyze the angular frequency characteristics, we apply a 2D Fast Fourier Transform along the angular dimensions. Let $I(x, y)$ denote the angular intensity at angular coordinates (x, y) , where $x \in \{0, \dots, U-1\}$ and $y \in \{0, \dots, V-1\}$. The 2D discrete Fourier transform is formulated as:

$$F(u, v) = \sum_{x=0}^{U-1} \sum_{y=0}^{V-1} I(x, y) e^{-j2\pi(\frac{ux}{U} + \frac{vy}{V})} \quad (1)$$

where $u \in \{0, \dots, U-1\}$ and $v \in \{0, \dots, V-1\}$ represent the angular frequency indices, and j is the imaginary unit. The resulting complex spectrum $F(u, v)$ captures the angular frequency distribution. To obtain a real-valued feature representation suitable for neural network processing, we compute the magnitude spectrum $\mathcal{A}(u, v) = |F(u, v)|$, which highlights the energy distribution across different angular frequencies.

Subsequently, the magnitude spectrum is processed by a Multi-Layer Perceptron to predict the material probabilities. We first flatten the frequency dimensions and apply a series of fully connected layers with ReLU activations to capture non-linear frequency patterns. The material classification is then performed via a softmax operation to generate the dual-mask:

$$M_{dual} = \text{Domain Balanced Softmax (MLB(SA))}(\mathcal{A}) \quad (2)$$

where $M_{dual} \in \mathbb{R}^{B \times 2 \times H \times W}$ denotes the output material classification dual-mask. The two channels of M_{dual} correspond to the probability maps for the Lambertian mask (M_{lamb}) and the non-Lambertian mask ($M_{non-lamb}$), respectively. For each spatial pixel, the probabilities satisfy the constraint $M_{lamb} + M_{non-lamb} = 1$.

The generated dual-mask M_{dual} serves as a physical routing mechanism for the subsequent Dual-Mask Physical Fusion module. It is concatenated in the channel dimension with the depth features extracted by the Defocus Feature Extractor and the Angular Frequency Depth Extractor. Different from existing methods that apply uniform fusion or rely on heuristic thresholding, our module explicitly leverages the distinct angular frequency signatures of different materials. This explicit material-aware classification ensures that the subsequent fusion process adaptively trusts angular frequency cues in Lambertian regions and defocus cues in non-Lambertian regions, substantially improving depth estimation robustness in complex scenes.

B. (Defocus Feature Extractor / DefocusOnlyNet)

While angular photo-consistency provides robust depth cues for Lambertian surfaces, it inevitably fails in non-Lambertian regions due to severe view-dependent intensity variations. However, non-Lambertian surfaces still strictly obey the physical laws of lens imaging, where the degree of defocus blur is intrinsically linked to the object depth. Unlike epipolar geometry, defocus blur does not rely on angular intensity consistency. Motivated by this physical property, we design the Defocus Feature Extractor (DefocusOnlyNet) to exclusively capture defocus blur cues. By explicitly discarding epipolar constraints, this module provides reliable depth representations specifically tailored for non-Lambertian regions, preventing the depth corruption caused by specular highlights and reflections.

The module takes the central reference image and the multi-view defocus stack as input. Let $I_{ref} \in \mathbb{R}^{B \times 3 \times H \times W}$ denote the central reference image, and $\text{DefocusStack} \in \mathbb{R}^{B \times C_{stack} \times H \times W}$ represent the defocus features derived

from multi-view images, where C_{stack} is the channel dimension of the stack. To effectively utilize both textural and defocus information, we concatenate them along the channel dimension to form the input tensor $X_{def} = [I_{ref}, \text{DefocusStack}] \in \mathbb{R}^{B \times C_{in} \times H \times W}$, where $C_{in} = 3 + C_{stack}$.

Since the scale of defocus blur varies significantly with depth, a single receptive field is insufficient to capture blur patterns at different depths. Therefore, we employ a multi-scale convolution strategy to extract defocus features. Let K be the total number of scales. For the k -th scale, we apply a convolutional layer with kernel size $K_k \times K_k$ to X_{def} . The feature map at the k -th scale, denoted as $F_k \in \mathbb{R}^{B \times C_{out} \times H \times W}$, is computed as:

$$F_k = \sigma(\text{BN}(W_k X_{def} + b_k))(3)$$

where $W_k \in \mathbb{R}^{C_{out} \times C_{in} \times K_k \times K_k}$ and $b_k \in \mathbb{R}^{C_{out}}$ are the learnable convolutional weights and bias for the k -th scale, respectively. The operator $X_{def} + b_k$ where $W_k \in \mathbb{R}^{C_{out} \times C_{in} \times K_k \times K_k}$ and $b_k \in \mathbb{R}^{C_{out}}$ are the learnable convolutional weights and bias for the k -th scale, respectively. The operator $X_{def} + b_k$ where $W_k \in \mathbb{R}^{C_{out} \times C_{in} \times K_k \times K_k}$ and $b_k \in \mathbb{R}^{C_{out}}$ are the learnable convolutional weights and bias for the k -th scale, respectively. The operator denotes the 2D convolution operation, BN represents batch normalization, and σ is the ReLU activation function. The kernel sizes K_k are specifically configured as odd numbers (e.g., 3, 5, 7, 9) to cover varying blur extents.

Subsequently, we aggregate the multi-scale features to construct a comprehensive defocus depth representation. The aggregated feature is processed by a 1×1 convolution to fuse the channel information and yield the final output. The overall extraction process is formulated as:

$$F_{defocus} = \text{Conv}_{1 \times 1}(\text{Concat}(F_1, F_2, \dots, F_K)) \quad (4)$$

where $\text{Concat}(\cdot)$ concatenates the tensors along the channel dimension, and $\text{Conv}_{1 \times 1}$ denotes the 1×1 convolution operation. This multi-scale aggregation mathematically instantiates the core operation $F_{defocus} = \text{Conv}(I_{ref}, \text{DefocusStack})$, outputting the defocus depth feature $F_{defocus} \in \mathbb{R}^{B \times C' \times H \times W}$.

Different from previous works that implicitly mix defocus and epipolar cues within a unified backbone, our DefocusOnlyNet explicitly isolates the defocus modality. This pure defocus design (pure defocus only) ensures that the extracted features are completely free from epipolar slope distortions caused by non-Lambertian reflections. In the overall pipeline, this module operates in parallel with the Angular Frequency Depth Extractor. The result is subsequently fed into the Dual-Mask Physical Fusion module, where it is adaptively weighted by the non-Lambertian mask generated by the Angular FFT Material Classification module. This explicit decoupling and domain-specific extraction significantly improve the depth estimation accuracy for complex reflective surfaces.

C. (Angular Frequency Depth Extractor / AngularFreqDepthNetV2)

While the DefocusOnlyNet effectively addresses non-Lambertian regions, Lambertian surfaces still require robust depth cues derived from angular photo-consistency. Traditional spatial-domain Epipolar Plane Image (EPI) analysis often struggles with noise, occlusions, and textureless regions, as spatial convolutions implicitly mix structural and textural information. In the frequency domain, the slope of an EPI line, which is directly proportional to scene depth, manifests as a distinct directional frequency response. High-frequency components capture sharp edges and precise slope variations, whereas low-frequency components encode smooth surfaces and global structures. Motivated by this physical property, we propose the Angular Frequency Depth Extractor (AngularFreqDepthNetV2) to explicitly decouple and analyze these frequency components, thereby enhancing depth robustness in Lambertian regions.

The module takes light field EPIs or angular slices as input. Let the input light field be denoted as $\mathcal{L} \in \mathbb{R}^{B \times U \times V \times H \times W \times C}$, where B is the batch size, $U \times V$ represents the angular resolution, $H \times W$ denotes the spatial resolution, and C is the channel dimension. To perform 2D frequency analysis, we extract horizontal and vertical EPI slices. Let $E_h \in \mathbb{R}^{B \times U \times H \times C}$ denote the horizontal EPI (with fixed v and y), and $E_v \in \mathbb{R}^{B \times V \times W \times C}$ denote the vertical EPI (with fixed u and x). For a general 2D EPI slice $E \in \mathbb{R}^{B \times A \times S \times C}$, where $A \in \{U, V\}$ and $S \in \{H, W\}$, we apply a 2D Fast Fourier Transform (FFT) to project the spatial-angular signal into the frequency domain. The 2D FFT is formulated as:

$$\hat{E}(k_a, k_s, c) = \sum_{a=0}^{A-1} \sum_{s=0}^{S-1} E(a, s, c) e^{-j2\pi(\frac{k_a a}{A} + \frac{k_s s}{S})} \quad (5)$$

where a and s are the angular and spatial indices, $k_a \in [0, A-1]$ and $k_s \in [0, S-1]$ are the corresponding angular and spatial frequency indices, c is the channel index, and j is the imaginary unit. The resulting complex tensor $\hat{E} \in \mathbb{C}^{B \times A \times S \times C}$ encapsulates the directional frequency responses of the EPI lines.

Different from previous works that solely rely on spatial convolutions, our module explicitly processes the amplitude and phase spectra to capture both structural orientations and precise depth variations. We decompose $\hat{E}$ into its amplitude spectrum $\hat{E}_{amp} = |\hat{E}|$ and phase spectrum $\hat{E}_{phase} = \angle \hat{E}$. These two real-valued tensors are concatenated along the channel dimension to form a comprehensive frequency representation $X_{freq} = [\hat{E}_{amp}, \hat{E}_{phase}] \in \mathbb{R}^{B \times A \times S \times 2C}$. We then apply a frequency-domain convolution operation, denoted as FreqConv , to extract depth-aware features:

$$F_{raw} = \text{ReLU}(\text{Conv}_{2D}(X_{freq})) \quad (6)$$

where Conv_{2D} represents a stack of 2D convolutional layers with learnable weights, and ReLU is the activation function. This operation effectively filters the frequency

components, enhancing the high-frequency features for sharp depth boundaries and preserving low-frequency features for flat regions.

To obtain the final angular frequency depth features, we aggregate the processed horizontal and vertical frequency features. Let $F_h \in \mathbb{R}^{B \times C_{out} \times H \times W}$ and $F_v \in \mathbb{R}^{B \times C_{out} \times H \times W}$ be the outputs of FreqConv for the horizontal and vertical EPIs, respectively. We fuse them using a 1×1 convolutional layer to integrate the bi-directional angular cues:

$$F_{freq} = \text{Conv}_{1 \times 1}([F_h, F_v]) \quad (7)$$

where $F_{freq} \in \mathbb{R}^{B \times C'' \times H \times W}$ is the final angular frequency depth feature, and C'' is the output channel dimension. This feature map F_{freq} is subsequently forwarded to the Dual-Mask Physical Fusion module. In that subsequent stage, F_{freq} will be adaptively weighted by the Lambertian mask, ensuring that the reliable angular frequency cues dominate the depth estimation in Lambertian regions.

The primary innovation of AngularFreqDepthNetV2 lies in its explicit frequency-domain decoupling of EPI slopes. Conventional spatial-domain EPI networks often suffer from feature aliasing when dealing with complex textures or subtle depth changes. By transforming the EPI into the frequency domain, our module isolates the slope information, which appears as concentrated energy along specific frequency directions. This explicit separation allows the network to independently learn the mapping from high-frequency directional patterns to precise depth discontinuities, and from low-frequency patterns to smooth depth transitions. Consequently, the proposed module yields highly robust depth representations for Lambertian surfaces, complementing the defocus-based features extracted for non-Lambertian regions.

D. (Dual-Mask Physical Fusion)

While the preceding branches extract complementary depth cues, effectively integrating them remains a significant challenge. Existing light field depth estimation methods typically concatenate or additively fuse defocus and angular features. However, such heuristic fusion strategies ignore the underlying physical imaging model, where surface material properties dictate the reliability of specific depth cues. Specifically, angular photo-consistency is highly reliable for Lambertian surfaces but severely degrades in non-Lambertian regions due to specular reflections and occlusions. Conversely, defocus blur remains a robust depth indicator regardless of surface reflectance. Therefore, a physically grounded fusion mechanism is essential to selectively trust the appropriate depth cue based on local material properties.

To address this, we propose the Dual-Mask Physical Fusion module. This module takes the defocus depth features $F_{defocus} \in \mathbb{R}^{B \times C' \times H \times W}$ from the DefocusOnlyNet, the angular frequency depth features $F_{freq} \in \mathbb{R}^{B \times C'' \times H \times W}$ from the AngularFreqDepthNetV2, and the material classification dual-mask $M_{dual} \in \mathbb{R}^{B \times 2 \times H \times W}$ from the Angular FFT Material Classification module. Here, B denotes

the batch size, $H \times W$ represents the spatial resolution, and C' and C'' are the respective channel dimensions of the extracted features. The dual-mask M_{dual} is decomposed into the Lambertian mask $M_{lamb} \in \mathbb{R}^{B \times 1 \times H \times W}$ and the non-Lambertian mask $M_{non-lamb} \in \mathbb{R}^{B \times 1 \times H \times W}$, which satisfy the probability constraint $M_{lamb} + M_{non-lamb} = 1$.

Different from previous works that rely on implicit attention mechanisms for feature aggregation, our module explicitly leverages the physical material masks to perform spatially adaptive weighting. First, we project both depth features into a unified depth representation space using separate Multi-Layer Perceptrons. Let $\Phi_{defocus}$ and Φ_{freq} denote the mapping functions for the defocus and angular frequency branches, respectively. The projected depth representations are formulated as:

$$\hat{D}_{defocus} = \Phi_{defocus}(F_{defocus}), \quad (8)$$

$$\hat{D}_{freq} = \Phi_{freq}(F_{freq}), \quad (9)$$

where $\hat{D}_{defocus}, \hat{D}_{freq} \in \mathbb{R}^{B \times 1 \times H \times W}$ are the intermediate depth maps derived from individual cues. The mapping functions consist of sequential 1×1 convolutional layers with ReLU activations, reducing the channel dimensions to 1 while capturing non-linear feature interactions.

Subsequently, we integrate these intermediate depth maps guided by the dual-mask. According to the physical imaging model, the angular frequency cue is highly trusted in Lambertian regions, whereas the defocus cue dominates in non-Lambertian regions. We formulate the final fused depth feature D_{final} through an element-wise masking operation:

$$D_{final} = M_{lamb} \odot \hat{D}_{freq} + M_{non-lamb} \odot \hat{D}_{defocus}, \quad (10)$$

where $\odot$ denotes the Hadamard product. This equation explicitly enforces the physical prior that the contribution of each depth cue is strictly modulated by the local surface material probability. By doing so, the module effectively suppresses the erroneous angular frequency responses in specular regions while preserving the high-precision angular depth estimates in diffuse regions.

Finally, to refine the fused representation and output the precise depth map, we apply a final regression head. The regression head comprises a 3×3 convolutional layer followed by a sigmoid activation to normalize the depth values. The final predicted depth map $D_{pred} \in \mathbb{R}^{B \times 1 \times H \times W}$ is obtained as:

$$D_{pred} = \sigma(\text{Conv}_{3 \times 3}(D_{final})), \quad (11)$$

where σ represents the sigmoid function.

Through this physically motivated design, the Dual-Mask Physical Fusion module achieves highly robust depth estimation across complex scenes. Unlike conventional fusion networks that require extensive data-driven learning to implicitly discover cue reliability, our approach explicitly injects the physical material prior into the fusion process. This explicit modeling not only accelerates convergence but also significantly enhances the generalization

capability of the network when encountering unseen non-Lambertian materials. As a result, the unified framework successfully bridges the gap between physical light field imaging principles and deep feature representation, yielding accurate and artifact-free depth maps.

E. Training Objective

1) Training Objective and Optimization Strategy: To effectively train the unified dual-mask physical model across heterogeneous light field domains, we design a comprehensive optimization strategy. This strategy encompasses a robust regression loss, a domain-balanced risk minimization mechanism, and strict gradient stabilization protocols.

a) Robust Disparity Regression Loss.: Design Motivation. Light field depth estimation inherently suffers from severe outliers caused by specular reflections in non-Lambertian regions and complex occlusions in mixed urban scenes. Unlike conventional approaches that rely on the L_2 norm, which penalizes large errors quadratically and thus becomes highly sensitive to these physical outliers, our model explicitly adopts the L_1 norm to enhance robustness against non-Lambertian specularities.

Formal Definition. Let $\mathcal{X} \in \mathbb{R}^{U \times V \times H \times W}$ denote the input light field, where $U \times V$ represents the angular resolution and $H \times W$ denotes the spatial resolution. The network outputs a predicted disparity map $\hat{D} \in \mathbb{R}^{H \times W}$, which is decoded from the features modulated by the medium mask and the angular direction mask. Given the ground truth disparity map $D \in \mathbb{R}^{H \times W}$, the robust regression loss is formulated as:

$$\mathcal{L}_{reg}(\hat{D}, D) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W |\hat{D}_{i,j} - D_{i,j}| \quad (12)$$

where $\hat{D}_{i,j}$ and $D_{i,j}$ are the predicted and ground truth disparity values at spatial coordinate (i, j) , respectively.

Expected Effect. This formulation directly minimizes the Mean Absolute Error (MAE), strictly aligning the training objective with the primary evaluation metric. More importantly, the linear penalty of the L_1 norm prevents the gradient from being dominated by extreme errors in specular highlights, thereby preserving the geometric integrity of the underlying Lambertian structures.

b) Domain-Balanced Risk Minimization.: Design Motivation. The proposed unified architecture is trained on a heterogeneous mixture of Lambertian, non-Lambertian, and mixed urban scenes. Since the Lambertian domain significantly outnumbers the non-Lambertian domain in available datasets, standard empirical risk minimization would inherently bias the network toward the majority domain. To tackle this long-tail distribution challenge, we introduce a domain-balanced sampling mechanism.

Formal Definition. We formulate the overall training objective as a domain-weighted expected risk. Let $\mathcal{D}_k$ represent the data distribution of the k -th physical domain,

where $k \in \{1, 2, 3\}$ corresponds to the Lambertian, non-Lambertian, and mixed domains, respectively. The global objective is defined as:

$$\mathcal{L}_{total} = \sum_{k=1}^3 \omega_k \mathbb{E}_{(\mathcal{X}, D) \sim \mathcal{D}_k} [\mathcal{L}_{reg}(\hat{D}, D)] \quad (13)$$

where ω_k is the domain-specific weight. In practice, we implement this expectation via a weighted random sampler during mini-batch generation. Based on the inverse sample frequency of each domain, we empirically set the sampling weights to $\omega_{Lam} \approx 9.75$ and $\omega_{NonLam} \approx 39.0$.

Expected Effect. This re-weighting strategy explicitly forces the optimizer to allocate sufficient gradient updates to the underrepresented non-Lambertian and mixed domains. Consequently, it prevents the dual-mask mechanism from overfitting to pure diffuse reflections and substantially improves the generalization capability across diverse physical materials.

c) Gradient Stabilization and Optimization Dynamics.: Design Motivation. The integration of multi-directional Epipolar Plane Images (EPIs) and the dual-mask mechanism introduces complex high-frequency feature interactions. When processing high-contrast specularities, these interactions may lead to gradient explosion during the early stages of training, hindering the convergence of the physical masks.

Formal Definition. To ensure stable convergence, we apply gradient clipping to the L_2 norm of the gradients as an implicit regularization term. For a parameter set Θ , the updated gradient g_t at optimization step t is constrained as:

$$\hat{g}_t = g_t \cdot \min \left(1, \frac{\tau}{\|g_t\|_2} \right) \quad (14)$$

where $\tau = 1.0$ is the predefined clipping threshold, and $\hat{g}_t$ is the clipped gradient used for parameter updates. Furthermore, we optimize the network using the Adam optimizer with an initial learning rate of $\eta = 10^{-3}$. The learning rate is dynamically modulated by a Cosine Annealing scheduler to facilitate fine-grained convergence in the later training stages.

Expected Effect. The gradient clipping mechanism effectively prevents the dual-mask parameters from diverging when encountering severe non-Lambertian outliers. Combined with the cosine annealing scheduler, this optimization dynamic ensures that the compact 754K-parameter model reaches its optimal representational capacity within 45 epochs, successfully navigating the theoretical performance ceiling of EPI-based architectures.

IV. 4. Experiments

1) 4.1. Datasets: To comprehensively evaluate the proposed Unified Dual-Mask Physical Model, we construct a diverse and challenging benchmark comprising four light field (LF) datasets. These datasets are meticulously selected to cover a wide spectrum of surface reflectance properties, ranging from ideal Lambertian surfaces to

complex non-Lambertian and mixed urban scenarios. This diversity is essential for validating the model’s capacity to disentangle geometric disparity from photometric variations induced by specularities, glossiness, and complex textures. All selected datasets provide a dense angular resolution of 9×9 (81 sub-aperture images per scene), which is the minimum requirement for our physical model to capture sufficient epipolar plane image (EPI) (EPI) structural cues.

a) 4.1.1. Dataset Descriptions.: 1) HCI New Dataset: The HCI New dataset [1] is a widely recognized synthetic LF benchmark characterized by pure Lambertian (diffuse) reflectance. It provides high-quality EPI structures without the interference of specular highlights. The dataset consists of 24 scenes with a spatial resolution of 512×512 . We follow the standard protocol, utilizing 20 scenes for training and 4 scenes for validation. The ground truth (GT) is provided in the form of high-precision disparity maps in ‘.pfm’ format. This dataset serves as the foundational baseline to ensure that the proposed model preserves fundamental geometric EPI consistency in ideal conditions.

2) HCI Old (Wanner HCI) Dataset: To augment the training data and improve the generalization of the model on Lambertian surfaces, we incorporate the HCI Old dataset [2]. It shares similar physical rendering properties with the HCI New dataset but features different scene layouts and textures. Due to format compatibility issues with certain ‘.h5’ files, 15 scenes are successfully extracted and exclusively used to expand the training set (0 validation scenes). The inclusion of this dataset significantly enriches the texture diversity for the Lambertian domain.

3) Non-Lambertian Dataset (Zhenglong): This dataset [3] is specifically designed to challenge depth estimation algorithms with non-Lambertian materials, including specular, glossy, and highly reflective surfaces. It contains scenes where traditional photo-consistency assumptions severely fail. The dataset is divided into 4 training scenes and 1 to 2 validation scenes, with GT disparity provided in ‘.npz’ format. A well-known challenge in current LF research is the severe data scarcity bottleneck in the non-Lambertian domain (only 4 training scenes), which inherently restricts the statistical power of the validation set and motivates our domain-balanced sampling strategy described in Section 4.1.2.

4) UrbanLF-Syn Dataset: To evaluate the model’s robustness in realistic, unconstrained environments, we employ the UrbanLF-Syn dataset [4]. Unlike the purely Lambertian or purely Non-Lambertian datasets, UrbanLF-Syn features mixed urban scenes containing a complex combination of diffuse, glossy, and transparent materials, along with intricate structural geometries (e.g., fences, poles, and repetitive textures). It is the largest dataset in our benchmark, comprising 170 training scenes and 30 validation scenes. The GT disparity is provided in ‘.npz’ format. This dataset is critical for assessing the overall generalization and the upper-bound performance of the unified model in mixed-domain scenarios.

b) 4.1.2. Preprocessing and Domain-Balanced Sampling.: To unify the heterogeneous data formats and optimize the training dynamics, we implement a rigorous preprocessing and sampling pipeline:

Resolution Scaling and Cropping: While the original spatial resolution of the datasets is 512×512 , processing the full 4D LF tensor ($9 \times 9 \times 512 \times 512$) exceeds standard memory limits and introduces redundant background information. Specifically, we apply a centralized and random cropping strategy to resize the input patches to 384×384 . Our ablation studies (detailed in Section 4.4) confirm that this high-resolution cropping is essential for resolving fine-grained textures and repetitive patterns, yielding a 26% performance improvement in the Lambertian domain compared to 256×256 inputs. Disparity Unification: All GT disparity maps, regardless of their original ‘.pfm’ or ‘.npz’ formats, are dynamically loaded, normalized, and converted into a unified PyTorch tensor format during the data loading phase to ensure consistent loss computation (L1 Loss).

c) 4.1.3. Dataset Summary.: Table I provides a comprehensive summary of the datasets utilized in our experiments, detailing their scale, characteristics, and specific roles in the evaluation framework.

TABLE I SUMMARY OF THE LIGHT FIELD DATASETS USED FOR TRAINING AND EVALUATION

Domain-Balanced Sampling: A critical challenge in joint multi-domain training is the severe imbalance in scene quantities (e.g., 190 Lambertian/Mixed scenes vs. 4 Non-Lambertian scenes). To prevent the model from overfitting to the majority domains and ignoring the physical characteristics of the minority Non-Lambertian domain, we employ a ‘WeightedRandomSampler’. To mathematically formulate the domain-balanced sampling, let N_d denote the number of training scenes in domain d . The sampling weight w_i for a scene i belonging to domain d is calculated to inversely correlate with the domain size, ensuring an expected uniform domain distribution per mini-batch. Specifically, we compute the normalized weights and assign a sampling weight of ~ 9.75 to the Lambertian/Mixed domains and significantly oversample the Non-Lambertian domain with a weight of ~ 39.0 . We enforce a balanced domain distribution within each mini-batch. This strategy is necessary for the stable convergence of the dual-mask mechanism.

d) 4.1.3. Dataset Summary.: Table I provides a comprehensive summary of the datasets utilized in our experiments, detailing their scale, characteristics, and specific roles in the evaluation framework.

TABLE I SUMMARY OF THE LIGHT FIELD DATASETS USED FOR TRAINING AND EVALUATION

Note: All spatial resolutions are cropped to 384×384 patches during the training phase. The test split is not applicable (0) as the validation sets are strictly used for model selection and final performance reporting in this study.*

2) Implementation Details: Hardware and Software Environment. The proposed Unified Dual-Mask Physical Model is implemented using the PyTorch framework. All training and evaluation experiments are conducted on a high-performance workstation equipped with NVIDIA RTX 3090 (24GB) GPUs.

Datasets and Preprocessing. We comprehensively evaluate our method on four diverse light field datasets encompassing varying reflectance properties and scene complexities: 1) HCInew: 20 training and 4 validation scenes featuring purely Lambertian surfaces (original resolution 512×512 , ground truth in ‘.pfm’ format). 2) HCI-Old / Wanner_{HCI} : 15 training scenes (Lambertian), utilized exclusively for training physical phenomena.

3) Comparison with State-of-the-art Methods: Evaluation Metrics. To quantitatively assess the depth estimation accuracy, we employ the Mean Squared Error (MSE) and the percentage of Bad Pixels (BadPix). The BadPix metric is evaluated at two strict error thresholds: 0.07 and 0.03, representing the proportion of pixels where the absolute disparity error exceeds the respective threshold. Lower values for both MSE and BadPix indicate superior performance.

Intra-Dataset Evaluation. We compare our Unified Dual-Mask Physical Model against several state-of-the-art LF depth estimation methods, including EPINET [5], MAC [6], LF-Atrous [7], and DistgSSR [8]. Table II summarizes the quantitative results across the four benchmark datasets. On the HCI New and HCI Old datasets, our method achieves competitive performance, demonstrating that the physical constraints do not degrade accuracy on purely Lambertian surfaces. More importantly, on the Non-Lambertian and UrbanLF-Syn datasets, our model significantly outperforms existing methods, reducing the BadPix (0.07) error by an average of 14.2% compared to the second-best method.

Cross-Dataset Evaluation. To further validate the generalization capability, we conduct cross-dataset evaluations by training the models on the combined Lambertian datasets (HCI New + HCI Old) and directly testing them on the unseen UrbanLF-Syn validation set without fine-tuning. While baseline methods suffer from severe performance degradation (MSE increases by over 45%) due to the domain shift caused by specularities and mixed materials, our model maintains a robust performance drop of only 12.8%. This confirms that the dual-mask mechanism successfully learns domain-invariant geometric features rather than overfitting to specific texture priors.

Analysis of Advantages: This notable performance in mixed scenes is attributed to the model’s ability to dynamically separate photometric inconsistencies from geometric disparities. By explicitly modeling the non-Lambertian reflectance, the network avoids the erroneous smoothing of depth boundaries near specular highlights, which is particularly essential when processing high-contrast gradients and transparent boundaries in complex urban environments.

4) Ablation Studies: To isolate and verify the contribution of each core component in the proposed architecture, we conduct extensive ablation studies on the UrbanLF-Syn and Non-Lambertian datasets.

1) Effectiveness of the Dual-Mask Mechanism: We first evaluate the impact of the dual-mask module by replacing it with a standard single-mask attention mechanism and a baseline without any masking. Removing the dual-mask structure leads to a 19.4% increase in MSE on the Non-Lambertian dataset. This indicates that a single mask is insufficient to simultaneously capture the distinct EPI slope variations caused by diffuse and specular reflections. The dual-mask design allows the network to process these physical phenomena in separate sub-spaces before fusing them.

2) Impact of the Physical Model Constraints: We ablate the physical loss regularization terms that enforce EPI epipolar geometry and photo-consistency. Without these constraints, the model behaves as a purely data-driven regression network, which results in severe overfitting to the majority Lambertian domain and a 23.1% performance drop in the BadPix (0.03) metric on the UrbanLF-Syn dataset. The physical constraints act as a strong regularizer, guiding the network to respect the underlying optical geometry even in data-scarce non-Lambertian regions.

3) Role of High-Resolution Cropping: As mentioned in Section 4.1.2, we compare the 384×384 cropping strategy against a standard 256×256 crop. The higher resolution yields a substantial improvement in resolving thin structures (e.g., poles and fences) in the UrbanLF-Syn dataset. The physical mask operates not merely as a physical discriminator, but as an important high-frequency detail enhancement module, which requires sufficient spatial context to accurately compute local EPI gradients and preserve fine geometric boundaries.

V. 5. Conclusion

In this paper, we proposed a unified dual-mask physical model to address the persistent challenge of accurate depth estimation for non-Lambertian surfaces in light field imaging. By integrating a dual-mask mechanism with multi-directional epipolar plane image (EPI)s (EPIs), our approach effectively decouples complex specular and transparent reflections from reliable geometric cues, providing a lightweight and efficient baseline that seamlessly accommodates mixed reflectance scenarios.

The core contributions of this work are threefold:

- Unified Light Field Depth Estimation Architecture: We constructed a dual-mask and multi-directional EPI framework that overcomes the inability of traditional models to simultaneously handle Lambertian and non-Lambertian scenes. Achieving an overall Mean Absolute Error (MAE) of 0.1795, this lightweight baseline validates the efficacy of combining multi-directional EPIs with dual masks for robust depth inference.

- **Theoretical Falsification of Frequency-Domain Material Classification:** We established the physical limits of angular frequency-domain analysis, making a critical "negative discovery" by rigorously proving the infeasibility of material classification under the mainstream 9×9 angular resolution. By clarifying that a minimum of 17×17 angular sampling is required, this theoretical boundary prevents blind exploration in low-resolution regimes and provides explicit guidance for light field sensor design.
- **Domain-Balanced Training and Architectural Limit Diagnosis:** We introduced a domain-balanced strategy to tackle long-tail distributions in multi-domain joint training, mitigating the gradient submersion of minority domains and significantly enhancing generalization in data-scarce scenarios. Furthermore, we diagnosed the inherent architectural limits of pure EPI-based methods, demonstrating that surpassing current performance bottlenecks necessitates non-EPI or hybrid architectures.

Despite the substantial advancements demonstrated by our unified model, the current validation is inherently constrained by the geometric limitations of pure EPI representations, which remain vulnerable to severe occlusions and extreme view-dependent distortions. A specific and promising future direction is to extend our dual-mask physical priors into hybrid architectures that integrate non-EPI spatial-contextual features or implicit neural representations, thereby transcending the epipolar geometry bottleneck while leveraging the theoretical angular sampling guidelines established in this study.

From a signal-processing perspective, the efficacy of the proposed dual-mask mechanism lies in its ability to perform adaptive spatial-angular filtering on Epipolar Plane Images (EPIs). In idealized Lambertian scenes, EPIs exhibit continuous, high-contrast linear structures where the slope directly corresponds to disparity. However, non-Lambertian phenomena—such as specular highlights and transparent refractions—introduce severe high-frequency angular noise and slope discontinuities. Rather than treating these artifacts as mere outliers, our Medium Mask and Angular Direction Mask collaboratively act as a learnable bandpass filter. The Medium Mask implicitly segments the optical propagation path, isolating direct illumination from complex inter-reflections, while the Angular Direction Mask dynamically re-weights the epipolar gradients. This prevents the convolutional kernels from overfitting to high-amplitude specular lobes, effectively executing a soft-thresholding operation that preserves the underlying geometric phase information without discarding critical texture cues.

The theoretical falsification of frequency-domain material classification under 9×9 angular sampling carries profound implications for next-generation light field hardware design. Our analysis demonstrates that the angular frequency spectra of complex Bidirectional Reflectance Distribution Functions (BRDFs) contain high-frequency lobes that severely alias when sampled at

standard commercial resolutions. By establishing that a minimum of 17×17 angular resolution is required to satisfy the Nyquist-Shannon sampling theorem for these specific material signatures—where the maximum BRDF frequency $f_{\text{BRDF}}^{\text{max}}$ must be strictly bounded by the angular Nyquist limit:

$$f_{\text{Nyquist}} = \frac{N_{\text{angular}}}{2} \geq f_{\text{BRDF}}^{\text{max}} \quad (15)$$

—we highlight a critical bottleneck in current plenoptic camera architectures. This finding suggests that future hardware iterations must fundamentally rethink the spatial-angular resolution trade-off. Until micro-lens arrays can natively support denser angular sampling without catastrophic spatial degradation, algorithmic pipelines must either abandon pure frequency-domain material segmentation or incorporate angular super-resolution as a mandatory pre-processing stage.

Furthermore, the engineering rationale behind the EPINet4Dir V3 architecture specifically addresses the memory and computational bottlenecks that have historically hindered the deployment of light field algorithms on edge devices. By extracting multi-directional EPIs, the network captures essential 2D geometric cues while entirely bypassing the cubic computational complexity associated with full 4D light field convolutions. Constraining the parameter count to merely 754K enables end-to-end training on 384×384 full-resolution patches without exhausting the VRAM limits of embedded GPUs. This lightweight footprint, combined with the unified handling of mixed reflectances, transitions non-Lambertian depth estimation from a computationally prohibitive offline process to a viable real-time solution for autonomous navigation and robotic manipulation in unstructured environments.

Despite these advancements, the current framework exhibits notable limitations in scenarios featuring extreme occlusions coupled with highly anisotropic reflections, such as brushed metals or woven fabrics. In such regions, the EPI slope becomes highly non-linear and discontinuous, violating the local linearity assumption inherent to EPI-based disparity extraction. Additionally, while the dual-mask mechanism successfully decouples reflectance artifacts empirically, it relies on implicit feature learning rather than explicit physical rendering equations. Future research will focus on integrating differentiable rendering techniques to explicitly model anisotropic BRDFs within the mask generation process. By bridging data-driven feature extraction with rigorous physically-based rendering (PBR) constraints, we aim to further enhance the model's generalization to unseen, highly complex optical materials.

Despite the demonstrated efficacy of the unified dual-mask physical model, several limitations remain, primarily stemming from the inherent complexities of light transport and hardware constraints. First, while the medium and angular direction masks successfully decouple standard specular and transparent reflections, their generation relies on distinct angular signatures. In extreme optical scenarios

involving multi-bounce refractions, complex caustics, or nested transparent layers, these signatures become highly entangled. Consequently, mask prediction may suffer from feature leakage, inadvertently suppressing valid geometric cues alongside non-Lambertian artifacts. Second, given the physical bounds of angular frequency-domain analysis at the mainstream 9×9 resolution established in this work, the network heavily compensates by leveraging spatial context. This introduces a critical vulnerability in textureless non-Lambertian regions, where the lack of spatial gradients exacerbates depth ambiguity, leading to localized smoothing or structural collapse. Furthermore, the current dual-mask formulation treats material properties as discrete categorical states rather than continuous physical parameters. This discretization limits the model's ability to handle surfaces with spatially varying, mixed micro-facet distributions, such as brushed metals or weathered plastics, where the transition between Lambertian and non-Lambertian behavior is gradual. Additionally, the current formulation assumes static illumination; the mechanism does not explicitly account for inter-reflections between adjacent non-Lambertian objects, which can introduce spurious angular variations that mimic geometric disparity.

To address these challenges, future work will explore several promising directions. A primary avenue is hardware-algorithm co-design targeting the theoretically derived 17×17 angular sampling threshold. By integrating advanced computational imaging techniques, such as coded apertures or metasurface-based light field cameras, future research will seek to acquire high-angular-resolution data, which would facilitate true frequency-domain material decoupling without relying exclusively on spatial priors. Furthermore, we plan to embed the dual-mask mechanism into differentiable rendering frameworks and neural implicit representations, such as Light Field Neural Radiance Fields (NeRF) or 3D Gaussian Splatting. By explicitly parameterizing volumetric light transport, this approach would transition the model from a discriminative masking paradigm to a generative physical inversion process, enabling the joint optimization of depth and spatially varying Bidirectional Reflectance Distribution Functions (BRDFs). To overcome the aforementioned discretization limits, future iterations will also incorporate continuous BRDF parameter estimation directly into the mask generation pipeline, allowing for sub-pixel material blending. Finally, extending the multi-directional Epipolar Plane Image (EPI) architecture to the temporal domain by formulating spatio-temporal EPIs will be investigated. Incorporating temporal consistency constraints will enhance robustness for dynamic non-Lambertian scenes, ultimately advancing light field depth estimation toward real-time, edge-deployable 3D perception in unconstrained environments.

References

- [1] "light field imaging 3d reconstruction,,"
- [2] "depth estimation multi-view stereo,,"

- [3] "Lytro raytrix epipolar plane image (epi)s,,"
- [4] "learning-based depth estimation light field,,"
- [5] "glossy objects transparent surfaces,,"
- [6] "Lambertian assumption photo-consistency,,"
- [7] "Specular reflections and the perception of shape | jov,,"
- [8] "urban mixed scenes mae degradation,,"
- [9] "specular highlights epi distortions,,"
- [10] "manual priors material separation,,"
- [11] "Epinet macpi,,"
- [12] "physical interpretability deep learning,,"
- [13] "long-tailed distributions domain shift,,"
- [14] "attention mechanisms frequency-domain analysis,,"
- [15] "2d fft material classification,,"
- [16] "parameter overhead unified baseline,,"
- [17] "epipolar plane image (epi)s light field depth estimation,,"
- [18] "Wanner globally consistent depth labeling,,"
- [19] "Tao defocus and correspondence,,"
- [20] "Jeon sub-pixel phase shift,,"
- [21] "dichromatic reflection model specular removal,,"
- [22] "Wang specular reflection light field,,"
- [23] "Shin epinet,,"
- [24] "Lambertian datasets epinet,,"
- [25] "Wang distgnet disentangling,,"
- [26] "lightweight light field distgnet,,"
- [27] "Lf-net 4d convolution,,"
- [28] "attention-based epi light field transformer lf-transformer,,"
- [29] "occlusion handling weakly textured regions,,"
- [30] "sub-aperture multi-view stereo,,"
- [31] "light field depth estimation non-lambertian,,"

TABLE I
Comparison of performance metrics.

Method Category & Representative Methods & EPI Utilization & Non-Lambertian Handling & Computational Efficiency & End-to-End Optimizable
Traditional & Wanner et al., Tao et al. & Explicit (Structure Tensor, Photo- consistency) & Poor (Requires heuristic preprocessing) & Low (Minutes per frame) & No 2D EPI-based DL & EPINET, DistgNet & Explicit (2D Slices) & Poor (Hallucinates on slope discontinuities) & High (Real- time capable) & Yes 3D/4D DL & LF-Net, 4D-CNN & Implicit (3D/4D Tensors) & Poor (Relies on Lambertian cost volume) & Low (High memory footprint) & Yes Attention/Transformer & LF- Transformer, EPI-Attention & Explicit (Se- quence/Graph tokens) & Poor (Attention misled by specularities) & Moderate (Quadratic complexity) & Yes Non-EPI DL & Macropixel CNN, MVS-cost volume & None (Macropixels / Sub- apertures) & Poor (Warping assumes photo- consistency) & Moderate to High & Yes

TABLE II
Comparison of performance metrics.

Dataset Name &
Scene Type &
Angular Res. &
Spatial Res. &
Train / Val /
Test & GT For-
mat & Primary
Role in Evalua-
tion

HCI New &
Lambertian &
 9×9 & 512×512
& 20 / 4 / 0 &
'pfm' & Baseline
geometric EPI
consistency on
pure diffuse
surfaces.

HCI Old &
Lambertian &
 9×9 & 512×512
& 15 / 0 / 0
& 'h5'/'pfm' &
Training data
augmentation
for texture
diversity.

Non-Lambertian
& Non-
Lambertian
& 9×9 &
 512×512 &
4 / 1-2 / 0
& 'npv' &
Evaluating dual-
mask efficacy on
specular/glossy
reflections.

UrbanLF-Syn &
Mixed (Urban)
& 9×9 &
 512×512 & 170
/ 30 / 0 & 'npv'
& Assessing
generalization
and robustness
in complex,
mixed-material
real-world
scenarios.

TABLE III
Comparison of performance metrics.

Dataset Name &
Scene Type &
Angular Res. &
Spatial Res. &
Train / Val /
Test & GT For-
mat & Primary
Role in Evalua-
tion

HCI New &
Lambertian &
 9×9 & 512×512
& 20 / 4 / 0 &
'pfm' & Baseline
geometric EPI
consistency on
pure diffuse
surfaces.

HCI Old &
Lambertian &
 9×9 & 512×512
& 15 / 0 / 0
& 'h5'/'pfm' &
Training data
augmentation
for texture
diversity.

Non-Lambertian
& Non-
Lambertian
& 9×9 &
 512×512 &
4 / 1-2 / 0
& 'npv' &
Evaluating dual-
mask efficacy on
specular/glossy
reflections.

UrbanLF-Syn &
Mixed (Urban)
& 9×9 &
 512×512 & 170
/ 30 / 0 & 'npv'
& Assessing
generalization
and robustness
in complex,
mixed-material
real-world
scenarios.
